

ASN Tracker — BLE Relay Manual

Connect external devices to receive live timer data over Bluetooth Low Energy

1. Enable Broadcast

1. Open the **ASN Tracker** app on the tablet.
2. Navigate to **Bluetooth Settings** (the Bluetooth icon in the top bar).
3. In the **Relay** section at the top, toggle **Broadcast** to ON.
4. The tablet will begin advertising as a BLE peripheral. Connected subscribers are shown below the toggle.

2. BLE Connection Details

Parameter	Value
Service UUID	E5A10001-F1A2-4B63-9F8C-D7B781E35E2A
Characteristic UUID	E5A10002-F1A2-4B63-9F8C-D7B781E35E2A
Properties	Notify + Read
Advertised Name	Device Bluetooth name (Android) / "ShotTimer Relay" (iOS)

To receive live data, connect to the service and **enable notifications** on the characteristic (write `0x01 0x00` to the CCCD descriptor `0x2902`). You can also read the characteristic to get the last transmitted value.

3. Protocol — Frame Format

All data is sent as BLE notifications using a PIE-compatible binary protocol:

```
[ 0xF8  0xF9 ]  control_code  data...  [ 0xF9  0xF8 ]
```

Field	Bytes	Description
Start sentinel	F8 F9	Marks the beginning of a frame
Control code	1 byte	Command type (see below)
Data	Variable	Command-specific payload
Stop sentinel	F9 F8	Marks the end of a frame

4. Commands

Command	Code	Data	Total bytes
Start	0x34	group (1 byte)	5
Stop	0x18	group (1 byte)	5
Hit	0x36	sec_hi, sec_lo, cs, hit#, group (5 bytes)	9

Time encoding: Seconds are a big-endian uint16 (`sec_hi << 8 | sec_lo`). The `cs` byte is centiseconds (0–99), representing the fractional part (`ms / 10`).

5. Examples

Event	Hex bytes
Timer started (group 0)	F8 F9 34 00 F9 F8
Timer stopped (group 0)	F8 F9 18 00 F9 F8
Hit #3 at 12.45s (group 0)	F8 F9 36 00 0C 2D 03 00 F9 F8

Note: In the hit example, 12 seconds = `0x00 0x0C` (big-endian), 45 centiseconds = `0x2D`, shot number 3 = `0x03`, group 0 = `0x00`.